GERSTING, JUDITH L. “Mathematical Structures for Computer Science: A Modern Approach to Discrete Mathematics”. W H Freeman & Co, 2014.

Chapter 2

Proofs, Induction, and Number Theory

CHAPTER OBJECTIVES

After studying this chapter, you will be able to:

- Attack the proofs of conjectures using the techniques of direct proof, proof by contraposition, and proof by contradiction.
- Recognize when a proof by induction is appropriate and carry out such a proof using either the first or second principle of induction.
- Mathematically prove the correctness of programs that use loop statements.
- Test whether a given positive integer is prime; if not, find its prime factorization.
- Work with number theoretic ideas of prime factorization, greatest common divisor, and the Euler phi function.

The nonprofit organization at which you volunteer has received donations of 792 bars of soap and 400 bottles of shampoo. You want to create packages to distribute to homeless shelters such that each package contains the same number of shampoo bottles and each package contains the same number of bars of soap.

Question: **How many packages can you create?**

This problem is solvable by trial and error, but it’s much easier to use an ancient algorithm that is discussed in this chapter.

First, however, we consider how to prove “real-world” arguments as opposed to the formal arguments of Chapter 1. It is helpful to have an arsenal of techniques for attacking a proof. Direct proof, proof by contraposition, and proof by contradiction are examined in Section 2.1. Many of the proofs given in this section are simple “number theory” results, that is, results about integers, such as “The product of two even integers is even.”

Section 2.2 concentrates on mathematical induction, a proof technique with particularly wide application in computer science. In Section 2.3 we see how, using induction, proof of correctness can be extended to cover looping statements. Finally, Section 2.4 explores some further number theory results, particularly concerning prime numbers.

SECTION 2.1 PROOF TECHNIQUES

Theorems and Informal Proofs

The formal arguments of Chapter 1 have the form $P \to Q$, where P and Q may represent compound statements. The point there was to prove that an argument is valid—true in all interpretations by nature of its internal form or structure, not because of its content or the meaning of its component parts. However, we often want to prove arguments that are not universally true but just true within some context. Meaning becomes important because we are discussing a particular subject—graph algorithms, or Boolean algebra, or compiler theory, or whatever—and we want to prove that if P is true in this context, then so is Q. If we can do this, then $P \to Q$ becomes a *theorem* about that subject. To prove a theorem about subject XXX, we can introduce facts about XXX into the proof; these facts act like additional hypotheses. Note that as we add more hypotheses, the universe of discourse shrinks; we are no longer considering universally valid arguments, only arguments that are true within the context in which the hypotheses hold.[1]

It may not be easy to recognize which subject-specific facts will be helpful or to arrange a sequence of steps that will logically lead from P to Q. Unfortunately, there is no formula for constructing proofs and no practical general algorithm or computer program for proving theorems. Experience is helpful, not only because you get better with practice, but also because a proof that works for one theorem can sometimes be modified to work for a new but similar theorem.

Theorems are often stated and proved in a somewhat less formal way than the propositional and predicate arguments of Chapter 1. For example, a theorem may express the fact that every object in the domain of interpretation (the subject matter under discussion) having property P also has property Q. The formal statement of the theorem would be $(\forall x)[P(x) \to Q(x)]$. But the theorem would be informally stated as $P(x) \to Q(x)$. If we can prove $P(x) \to Q(x)$ where x is treated as an arbitrary element of the domain, universal generalization would then give $(\forall x)[P(x) \to Q(x)]$.

As another example, we may know that all objects in the domain have some property; that is, something of the form $(\forall x)P(x)$ can be considered as a subject-specific fact. An informal proof might proceed by saying, "Let x be any element of the domain. Then x has property P." (Formally, we are making use of universal instantiation to get $P(x)$ from $(\forall x)P(x)$.)

Similarly, proofs are usually not written a step at a time with formal justifications for each step. Instead, the important steps and their rationale are outlined in narrative form. Such a narrative, however, can be translated into a formal proof if required. In fact, the value of a formal proof is that it serves as a sort of insurance—if a narrative proof *cannot* be translated into a formal proof, it should be viewed with great suspicion.

[1]In the world of "pure predicate logic," which is a correct and complete formal system, every true (valid) argument is provable. But in these more restrictive contexts, not everything that is "true" is necessarily provable, no matter how clever we are in adding additional hypotheses or "axioms." In other words, these systems may not be complete. At the age of 25, the German logician Kurt Gödel proved in 1931 that, using reasonable hypotheses, even elementary arithmetic is an incomplete system. This result shocked the mathematical community of the time, which had been depending on axiomatic systems since the days of Euclid.

To Prove or Not to Prove

A textbook will often say, "Prove the following theorem," and the reader will know that the theorem is true; furthermore, it is probably stated in its most polished form. But suppose you are doing research in some subject. You observe a number of cases in which whenever P is true, Q is also true. On the basis of these experiences, you may formulate a conjecture: $P \to Q$. The more cases you find where Q follows from P, the more confident you are in your conjecture. This process illustrates **inductive reasoning**, drawing a conclusion based on experience.

No matter how reasonable the conjecture sounds, however, you will not be satisfied until you have applied **deductive reasoning** to it as well. In this process, you try to verify the truth or falsity of your conjecture. You produce a proof of $P \to Q$ (thus making it a theorem), or else you find a **counterexample** that disproves the conjecture, a case in which P is true but Q is false. (We were using deductive reasoning in predicate logic when we either proved that a wff was valid or found an interpretation in which the wff was false.)

REMINDER

One counterexample is enough to disprove a conjecture.

If you are simply presented with a conjecture, it may be difficult to decide which of the two approaches you should try—to prove the conjecture or to disprove it! A single counterexample to a conjecture is sufficient to disprove it. Of course, merely hunting for a counterexample and being unsuccessful does not constitute a proof that the conjecture is true.

EXAMPLE 1

For a positive integer n, ***n* factorial** is defined as $n(n-1)(n-2)\cdots 1$, and is denoted by $n!$. Prove or disprove the conjecture, "For every positive integer n, $n! \le n^2$."

Let's begin by testing some cases:

n	$n!$	n^2	$n! \le n^2$
1	1	1	yes
2	2	4	yes
3	6	9	yes

So far, this conjecture seems to be looking good. But for the next case,

n	$n!$	n^2	$n! \le n^2$
4	24	16	no

we have found a counterexample. The fact that the conjecture is true for $n = 1, 2$, and 3 does nothing to prove the conjecture, but the single case $n = 4$ is enough to disprove it.

PRACTICE 1 Provide counterexamples to the following conjectures.

a. All animals living in the ocean are fish.
b. Every integer less than 10 is bigger than 5.

If a counterexample is not forthcoming, perhaps the conjecture is true and we should try to prove it. What techniques can we use to try to do this? For the rest of this section, we'll examine various methods of attacking a proof.

Exhaustive Proof

While "disproof by counterexample" always works, "proof by example" seldom does. The one exception to this situation occurs when the conjecture is an assertion about a finite collection. In this case, the conjecture can be proved true by showing that it is true for each member of the collection. **Proof by exhaustion** means that all possible cases have been exhausted, although it often means that the person doing the proof is exhausted as well!

EXAMPLE 2 Prove the conjecture, "If an integer between 1 and 20 is divisible by 6, then it is also divisible by 3." ("Divisible by 6," means, "evenly divisible by 6," that is, the number is an integral multiple of 6.)

Because there is only a finite number of cases, the conjecture can be proved by simply showing it to be true for all the integers between 1 and 20. Table 2.1 is the proof.

TABLE 2.1

Number	Divisible by 6	Divisible by 3
1	no	
2	no	
3	no	
4	no	
5	no	
6	yes: $6 = 1 \times 6$	yes: $6 = 2 \times 3$
7	no	
8	no	
9	no	
10	no	
11	no	
12	yes: $12 = 2 \times 6$	yes: $12 = 4 \times 3$
13	no	
14	no	
15	no	
16	no	
17	no	
18	yes: $18 = 3 \times 6$	yes: $18 = 6 \times 3$
19	no	
20	no	

EXAMPLE 3 Prove the conjecture, "It is not possible to trace all the lines in Figure 2.1 without lifting your pencil and without retracing any lines."

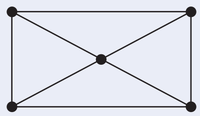


Figure 2.1

There is only a finite number of different ways to trace the lines in the figure. By careful bookkeeping, each of the possibilities can be attempted, and each will fail. In Chapter 7, we will learn a much less tedious way to solve this problem than proof by exhaustion.

PRACTICE 2

a. Prove the conjecture "For any positive integer less than or equal to 5, the square of the integer is less than or equal to the sum of 10 plus 5 times the integer."
b. Disprove the conjecture "For any positive integer, the square of the integer is less than or equal to the sum of 10 plus 5 times the integer."

Direct Proof

In general (where exhaustive proof won't work), how can you prove that $P \to Q$ is true? The obvious approach is the **direct proof**—assume the hypothesis P and deduce the conclusion Q. A formal proof would require a proof sequence leading from P to Q.

Example 4 shows a formal proof that if two numbers are even (that's the hypothesis P), then their product is even (that's the conclusion Q). Recall that an **even number** is a number that is an integral multiple of 2, for example, 18 is even because $18 = 2(9)$. An **odd number** is 1 more than an integral multiple of 2, for example, $19 = 2(9) + 1$.

EXAMPLE 4 Consider the conjecture

$$x \text{ is an even integer} \wedge y \text{ is an even integer} \to \text{the product } xy \text{ is an even integer}$$

A complete formal proof sequence might look like the following:

1. x is an even integer $\wedge$ y is an even integer	hyp
2. $(\forall x)[x$ is even integer $\to$ $(\exists k)(k$ an integer $\wedge$ $x = 2k)]$	number fact (definition of even integer)
3. x is even integer $\to$ $(\exists k)(k$ an integer $\wedge$ $x = 2k)$	2, ui
4. y is even integer $\to$ $(\exists k)(k$ an integer $\wedge$ $y = 2k)$	2, ui
5. x is an even integer	1, sim
6. $(\exists k)(k$ is an integer $\wedge$ $x = 2k)$	3, 5, mp
7. m is an integer $\wedge$ $x = 2m$	6, ei
8. y is an even integer	1, sim
9. $(\exists k)(k$ an integer $\wedge$ $y = 2k)$	4, 8, mp
10. n is an integer and $y = 2n$	9, ei
11. $x = 2m$	7, sim
12. $y = 2n$	10, sim

13. $xy = (2m)(2n)$	11, 12, substitution of equals
14. $xy = 2(2mn)$	13, multiplication fact
15. m is an integer	7, sim
16. n is an integer	10, sim
17. $2mn$ is an integer	15, 16, number fact
18. $xy = 2(2mn) \wedge 2mn$ is an integer	14, 17, con
19. $(\exists k)(k$ an integer $\wedge\, xy = 2k)$	18, eg
20. $(\forall x)((\exists k)(k$ an integer $\wedge\, x = 2k) \rightarrow$ x is even integer)	number fact (definition of even integer)
21. $(\exists k)(k$ an integer $\wedge\, xy = 2k) \rightarrow$ xy is even integer	20, ui
22. xy is an even integer	19, 21, mp

It is understood that x and y are arbitrary, but this could be stated by expressing the conjecture as

$$(\forall x)(\forall y)(x \text{ is an even integer} \wedge y \text{ is an even integer} \rightarrow \text{the product } xy \text{ is an even integer})$$

Universal generalization can be applied twice to the result we already have in order to put the universal quantifiers on the front.

We'll never again do a proof like the one in Example 4, and you won't have to either! A much more informal proof would be perfectly acceptable in most circumstances.

EXAMPLE 5

Following is an informal direct proof that the product of two even integers is even.

Let $x = 2m$ and $y = 2n$, where m and n are integers. Then $xy = (2m)(2n) = 2(2mn)$, where $2mn$ is an integer. Thus xy has the form $2k$, where k is an integer, and xy is therefore even.

Notice that we set $x = 2m$ for some integer m (the definition of an even number), but we set $y = 2n$. In the formal proof of Example 4, the restriction on the use of *ei* required that we use a different multiple of 2 for y than we used for x. Informally, if we were also to set $y = 2m$, we would be saying that x and y are the same integers, which is a very special case.

The proof in Example 5 does not explicitly state the hypothesis (that x and y are even), and it makes implicit use of the definition of an even integer. Even in informal proofs, however, it is important to identify the hypothesis and the conclusion, not just what they are in words but what they really mean, by applying appropriate definitions. If we do not clearly understand what we have (the hypothesis) or what we want (the conclusion), we cannot hope to build a bridge from one to the other. That's why it is important to know definitions.

PRACTICE 3 Give a direct proof (informal) of the theorem "If an integer is divisible by 6, then twice that integer is divisible by 4."

Contraposition

If you have tried diligently but failed to produce a direct proof of your conjecture $P \to Q$, and you still feel that the conjecture is true, you might try some variants on the direct proof technique. If you can prove the theorem $Q' \to P'$, you can conclude $P \to Q$ by making use of the tautology $(Q' \to P') \to (P \to Q)$. $Q' \to P'$ is the **contrapositive** of $P \to Q$, and the technique of proving $P \to Q$ by doing a direct proof of $Q' \to P'$ is called **proof by contraposition**. (The contraposition rule of inference in propositional logic (Table 1.14) says that $P \to Q$ can be derived from $Q' \to P'$.)

EXAMPLE 6 Prove that if the square of an integer is odd, then the integer must be odd.

The conjecture is n^2 odd $\to n$ odd. We do a proof by contraposition, and prove n even $\to n^2$ even. Let n be even. Then $n^2 = n(n)$ is even by Example 5.

EXAMPLE 7 Prove that if $n + 1$ separate passwords are issued to n students, then some student gets ≥ 2 passwords.

Here the conclusion Q has the form $(\exists x)R(x)$, so Q' is $[(\exists x)R(x)]'$, which is equivalent to $(\forall x)[R(x)]'$. The contrapositive $Q' \to P'$ is, "If every student gets < 2 passwords, then it is false that $n + 1$ passwords were issued." Suppose every student has < 2 passwords; then every one of the n students has at most 1 password. The total number of passwords issued is at most n, not $n + 1$, so it is false that $n + 1$ passwords were issued.

Example 7 is an illustration of the pigeonhole principle, which we will see in Chapter 4.

PRACTICE 4 Write the contrapositive of each statement in Practice 5 of Chapter 1.

Practice 7 of Chapter 1 showed that the wffs $A \to B$ and $B \to A$ are not equivalent. $B \to A$ is the **converse** of $A \to B$. If an implication is true, its converse may be true or false. Therefore, you cannot prove $P \to Q$ by looking at $Q \to P$.

EXAMPLE 8 The implication "If $a > 5$, then $a > 2$" is true, but its converse, "If $a > 2$, then $a > 5$," is false.

PRACTICE 5 Write the converse of each statement in Practice 5 of Chapter 1. ■

REMINDER

"If and only if" requires two proofs, one in each direction.

Theorems are often stated in the form "P if and only if Q," meaning P if Q and P only if Q, or $Q \to P$ and $P \to Q$. To prove such a theorem, you must prove both an implication and its converse. Again, the truth of one does not imply the truth of the other.

EXAMPLE 9 Prove that the product xy is odd if and only if both x and y are odd integers.

We first prove that if x and y are odd, so is xy. A direct proof will work. Suppose that both x and y are odd. Then $x = 2n + 1$ and $y = 2m + 1$, where m and n are integers. Then $xy = (2n + 1)(2m + 1) = 4nm + 2m + 2n + 1 = 2(2nm + m + n) + 1$. This has the form $2k + 1$, where k is an integer, so xy is odd.

Next we prove that if xy is odd, both x and y must be odd, or

$$xy \text{ odd} \to x \text{ odd and } y \text{ odd}$$

A direct proof would begin with the hypothesis that xy is odd, which leaves us little more to say. A proof by contraposition works well because we'll get more useful information as hypotheses. So we will prove

$$(x \text{ odd and } y \text{ odd})' \to (xy \text{ odd})'$$

By De Morgan's law $(A \wedge B)' \Leftrightarrow A' \vee B'$, we see that this can be written as

$$x \text{ even or } y \text{ even} \to xy \text{ even} \tag{1}$$

The hypothesis "x even or y even" breaks down into three cases. We consider each case in turn.

1. x even, y odd: Here $x = 2m$, $y = 2n + 1$, and then $xy = (2m)(2n + 1) = 2(2mn + m)$, which is even.
2. x odd, y even: This works just like case 1.
3. x even, y even: Then xy is even by Example 5.

This completes the proof of (1) and thus of the theorem. ●

The second part of the proof of Example 9 uses **proof by cases**, a form of exhaustive proof. It involves identifying all the possible cases consistent with the given information and then proving each case separately.

Contradiction

In addition to direct proof and proof by contraposition, you might use the technique of **proof by contradiction**. (Proof by contradiction is sometimes called *indirect proof*, but this term more properly means any argument that is not a direct proof.) As we did in Chapter 1, we will let 0 stand for any contradiction, that is, any wff whose truth value is always false. ($A \wedge A'$ would be such a wff.) Once more, suppose you are trying to prove $P \to Q$. By constructing a truth table, we see that

$$(P \wedge Q' \to 0) \to (P \to Q)$$

is a tautology, so to prove the theorem $P \to Q$, it is sufficient to prove $P \wedge Q' \to 0$. Therefore, in a proof by contradiction you assume that both the hypothesis and the negation of the conclusion are true and then try to deduce some contradiction from these assumptions.

EXAMPLE 10 Let's use proof by contradiction on the statement, "If a number added to itself gives itself, then the number is 0." Let x represent any number. The hypothesis is $x + x = x$ and the conclusion is $x = 0$. To do a proof by contradiction, assume $x + x = x$ and $x \neq 0$. Then $2x = x$ and $x \neq 0$. Because $x \neq 0$, we can divide both sides of the equation $2x = x$ by x and arrive at the contradiction $2 = 1$. Hence, $(x + x = x) \to (x = 0)$. ●

REMINDER

To prove that something is not true, try proof by contradiction.

Example 10 notwithstanding, a proof by contradiction most immediately comes to mind when you want to prove that something is *not* true. It's hard to prove that something *is not true*; it's much easier to *assume it is true* and obtain a contradiction.

EXAMPLE 11 A well-known proof by contradiction shows that $\sqrt{2}$ is not a rational number. Recall that a **rational number** is one that can be written in the form p/q where p and q are integers, $q \neq 0$, and p and q have no common factors (other than ± 1).

Let us assume that $\sqrt{2}$ is rational. Then $\sqrt{2} = p/q$, and $2 = p^2/q^2$, or $2q^2 = p^2$. Then 2 divides p^2, so—because 2 is itself indivisible—2 must divide p. This means that 2 is a factor of p; hence 4 is a factor of p^2, and the equation $2q^2 = p^2$ can be written as $2q^2 = 4x$, or $q^2 = 2x$. We see from this equation that 2 divides q^2 and hence 2 divides q. At this point, 2 is a factor of q and a factor of p, which contradicts the statement that p and q have no common factors. Therefore $\sqrt{2}$ is not rational. ●

The proof of Example 11 involves more than just algebraic manipulations. It is often necessary to use lots of words in a proof.

PRACTICE 6 Prove by contradiction that the product of odd integers is not even. (We did a direct proof of an equivalent statement in Example 9.) ■

Proof by contradiction can be a valuable technique, but it is easy to think we have done a proof by contradiction when we really haven't. For example, suppose we assume $P \wedge Q'$ and are able to deduce Q without using the assumption Q'. Then we assert $Q \wedge Q'$ as a contradiction. What really happened here is a direct proof of $P \to Q$, and the proof should be rewritten in this form. Thus in Example 10, we could assume $x + x = x$ and $x \neq 0$, as before. Then we could argue that from

$x + x = x$ we get $2x = x$ and, after subtracting x from both sides, $x = 0$. We then have $x = 0$ and $x \neq 0$, a contradiction. However, in this argument we never made use of the assumption $x \neq 0$; we actually proved directly that $x + x = x$ implies $x = 0$.

Another misleading claim of proof by contradiction occurs when we assume $P \wedge Q'$ and are able to deduce P' without using the assumption P. Then we assert $P \wedge P'$ as a contradiction. What really happened here is a direct proof of $Q' \rightarrow P'$, and we have constructed a proof by contraposition, not a proof by contradiction. In both this case and the previous one, it is not that the proofs are wrong, just that they are not proofs by contradiction.

Table 2.2 summarizes useful proof techniques we have discussed so far.

TABLE 2.2

Proof Technique	**Approach to Prove $P \rightarrow Q$**	**Remarks**
Exhaustive proof	Demonstrate $P \rightarrow Q$ for all possible cases.	May be used only to prove a finite number of cases.
Direct proof	Assume P, deduce Q.	The standard approach—usually the thing to try.
Proof by contraposition	Assume Q', deduce P'.	Use this if Q' as a hypothesis seems to give more ammunition than P would.
Proof by contradiction	Assume $P \wedge Q'$, deduce a contradiction.	Use this when Q says something is not true.

Serendipity

Serendipity means a fortuitous happening, or good luck. While this isn't really a general proof technique, some of the most interesting proofs come from clever observations that we can admire, even if we would never have thought of them ourselves. We'll look at two such proofs, just for fun.

EXAMPLE 12 A tennis tournament has 342 players. A single match involves 2 players. The winner of a match plays the winner of a match in the next round, while losers are eliminated from the tournament. The 2 players who have won all previous rounds play in the final game, and the winner wins the tournament. Prove that the total number of matches to be played is 341.

The hard way to prove this result is to compute $342/2 = 171$ to get the number of matches in the first round, resulting in 171 winners to go on to the second round. For the second round, $171/2 = 85$ plus 1 left over; there are 85 matches and 85 winners, plus the 1 left over, to go on to the third round. The third round has $86/2 = 43$ matches, and so forth. The total number of matches is the sum of $171 + 85 + 43 + \cdots$.

The clever observation is to note that each match results in exactly 1 loser, so there must be the same number of matches as losers in the tournament. Because there is only 1 winner, there are 341 losers and therefore 341 matches. ●

EXAMPLE 13 A standard 64-square checkerboard is arranged in 8 rows of 8 squares each. Adjacent squares are alternating colors of red and black. A set of 32 1×2 tiles, each covering 2 squares, will cover the board completely (4 tiles per row, 8 rows). Prove that if the squares at diagonally opposite corners of the checkerboard are removed, the remaining board cannot be covered with 31 tiles.

The hard way to prove this result is to try all possibilities with 31 tiles and see that they all fail. The clever observation is to note that opposing corners are the same color, so the checkerboard with the corners removed has two less squares of one color than of the other. Each tile covers one square of each color, so any set of tiles must cover an equal number of squares of each color and cannot cover the board with the corners removed.

Common Definitions

Many of the examples in this section and many of the exercises that follow involve elementary **number theory**, that is, results about integers. It's useful to work in number theory when first starting to construct proofs because many properties of integers, such as what it means to be an even number, are already familiar. The following definitions may be helpful in working some of these exercises.

- A **perfect square** is an integer n such that $n = k^2$ for some integer k.
- A **prime number** is an integer $n > 1$ such that n is not divisible by any integers other than 1 and n.
- A **composite number** n is a nonprime integer; that is, $n = ab$ where a and b are integers with $1 < a < n$ and $1 < b < n$.
- For two numbers x and y, $x < y$ means $y - x > 0$.
- For two integers n and m, n **divides** m, $n \mid m$, means that m is divisible by n—that is, $m = k(n)$ for some integer k.
- The **absolute value** of a number x, $|x|$, is x if $x \geq 0$ and is $-x$ if $x < 0$.

SECTION 2.1 REVIEW

TECHNIQUES

- Look for a counterexample.
- Construct direct proofs, proofs by contraposition, and proofs by contradiction.

MAIN IDEAS

- Inductive reasoning is used to formulate a conjecture based on experience. Deductive reasoning is used either to refute a conjecture by finding a counterexample or to prove a conjecture.
- In proving a conjecture about some subject, facts about that subject can be used.
- Under the right circumstances, proof by contraposition or contradiction may work better than a direct proof.

EXERCISES 2.1

1. Write the contrapositive of each statement in Exercise 5 of Section 1.1.
2. Write the converse of each statement in Exercise 5 of Section 1.1.

3. Provide counterexamples to the following statements.
   a. Every geometric figure with four right angles is a square.
   b. If a real number is not positive, then it must be negative.
   c. All people with red hair have green eyes or are tall.
   d. All people with red hair have green eyes and are tall.
4. Provide counterexamples to the following statements.
   a. If a and b are integers where $a \mid b$ and $b \mid a$, then $a = b$.
   b. If $n^2 > 0$ then $n > 0$.
   c. If n is an even number, then $n^2 + 1$ is prime.
   d. If n is a positive integer, then $n^3 > n!$.
5. Provide a counterexample to the following statement: The number n is an odd integer if and only if $3n + 5$ is an even integer.
6. Provide a counterexample to the following statement: The number n is an even integer if and only if $3n + 2$ is an even integer.
7. a. Find two even integers whose sum is not a multiple of 4.
   b. What is wrong with the following "proof" that the sum of two even numbers is a multiple of 4?

   Let x and y be even numbers. Then $x = 2m$ and $y = 2m$, where m is an integer, so $x + y = 2m + 2m = 4m$, which is an integral multiple of 4.
8. a. Find an example of an odd number x and an even number y such that $x - y = 7$.
   b. What is wrong with the following "proof" that an odd number minus an even number is always 1?

   Let x be odd and y be even. Then $x = 2m + 1$, $y = 2m$, where m is an integer, and $x - y = 2m + 1 - 2m = 1$.

For Exercises 9–46, prove the given statement.

9. If $n = 25$, 100, or 169, then n is a perfect square and is a sum of two perfect squares.
10. If n is an even integer, $4 \le n \le 12$, then n is a sum of two prime numbers.
11. For any positive integer n less than or equal to 3, $n! < 2^n$.
12. For $2 \le n \le 4$, $n^2 \ge 2^n$.
13. The sum of two even integers is even (do a direct proof).
14. The sum of two even integers is even (do a proof by contradiction).
15. The sum of two odd integers is even.
16. The sum of an even integer and an odd integer is odd.
17. An odd integer minus an even integer is odd.
18. If n is an even integer, then $n^2 - 1$ is odd.
19. The product of any two consecutive integers is even.
20. The sum of an integer and its square is even.
21. The square of an even number is divisible by 4.
22. For every integer n, the number $3(n^2 + 2n + 3) - 2n^2$ is a perfect square.
23. If a number x is positive, so is $x + 1$ (do a proof by contraposition).
24. If n is an odd integer, then it is the difference of two perfect squares.
25. The number n is an odd integer if and only if $3n + 5 = 6k + 8$ for some integer k.
26. The number n is an even integer if and only if $3n + 2 = 6k + 2$ for some integer k.

27. For x and y positive numbers, $x < y$ if and only if $x^2 < y^2$.
28. If $x^2 + 2x - 3 = 0$, then $x \neq 2$.
29. If n is an even prime number, then $n = 2$.
30. The sum of three consecutive integers is divisible by 3.
31. If two integers are each divisible by some integer n, then their sum is divisible by n.
32. If the product of two integers is not divisible by an integer n, then neither integer is divisible by n.
33. If n, m, and p are integers and $n \mid m$ and $m \mid p$, then $n \mid p$.
34. If n, m, p, and q are integers and $n \mid p$ and $m \mid q$, then $nm \mid pq$.
35. The square of an odd integer equals $8k + 1$ for some integer k.
36. The sum of the squares of two odd integers cannot be a perfect square. (*Hint:* Use Exercise 35.)
37. The product of the squares of two integers is a perfect square.
38. The difference of two consecutive cubes is odd.
39. For any two numbers x and y, $|x + y| \leq |x| + |y|$.
40. For any two numbers x and y, $|xy| = |x||y|$.
41. The value A is the average of the n numbers $x_1, x_2, \ldots, x_n$. Prove that at least one of $x_1, x_2, \ldots, x_n$ is greater than or equal to A.
42. Suppose you were to use the steps of Example 11 to attempt to prove that $\sqrt{4}$ is not a rational number. At what point would the proof not be valid?
43. Prove that $\sqrt{3}$ is not a rational number.
44. Prove that $\sqrt{5}$ is not a rational number.
45. Prove that $\sqrt[3]{2}$ is not a rational number.
46. Prove that $\log_2 5$ is not a rational number ($\log_2 5 = x$ means $2^x = 5$).

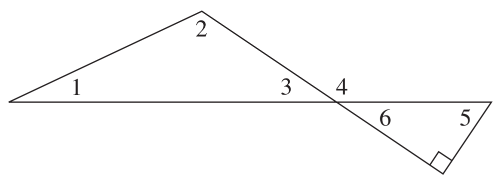
For Exercises 47–72, prove or disprove the given statement.

47. 0 is an even number.
48. 91 is a composite number.
49. 297 is a composite number.
50. 83 is a composite number.
51. The difference between two odd integers is odd.
52. The difference between two even integers is even.
53. The product of any three consecutive integers is even.
54. The sum of any three consecutive integers is even.
55. The sum of an integer and its cube is even.
56. The number n is an even integer if and only if $n^3 + 13$ is odd.
57. The product of an integer and its square is even.
58. Any positive integer can be written as the sum of the squares of two integers.
59. The sum of the square of an odd integer and the square of an even integer is odd.
60. If n is a positive integer that is a perfect square, then $n + 2$ is not a perfect square.
61. For a positive integer n, $n + \dfrac{1}{n} \geq 2$.

62. If n, m, and p are integers and $n \mid mp$, then $n \mid m$ or $n \mid p$.
63. For every prime number n, $n + 4$ is prime.
64. For every positive integer n, $n^2 + n + 3$ is not prime.
65. For n a positive integer, $n > 2$, $n^2 - 1$ is not prime.
66. For every positive integer n, $2^n + 1$ is prime.
67. For every positive integer n, $n^2 + n + 1$ is prime.
68. For n an even integer, $n > 2$, $2^n - 1$ is not prime.
69. The sum of two rational numbers is rational.
70. The product of two rational numbers is rational.
71. The product of two irrational numbers is irrational.
72. The sum of a rational number and an irrational number is irrational.

For Exercises 73–75, use the following facts from geometry and the accompanying figure.

- The interior angles of a triangle sum to 180°.
- Vertical angles (opposite angles formed when two lines intersect) are the same size.
- A straight angle is 180°.
- A right angle is 90°.



73. Prove that the measure of angle 5 plus the measure of angle 3 is 90°.
74. Prove that the measure of angle 4 is the sum of the measures of angles 1 and 2.
75. If angle 1 and angle 5 are the same size, then angle 2 is a right angle.
76. Prove that the sum of the integers from 1 through 100 is 5050. (*Hint:* Instead of actually adding all the numbers, try to make the same clever observation that the German mathematician Karl Friederick Gauss [1777–1855] made as a schoolchild: Group the numbers into pairs, using 1 and 100, 2 and 99, etc.)

SECTION 2.2 INDUCTION

First Principle of Induction

There is one final proof technique especially useful in computer science. To illustrate how the technique works, imagine that you are climbing an infinitely high ladder. How do you know whether you will be able to reach an arbitrarily high rung? Suppose we make the following two assertions about your climbing abilities:

1. You can reach the first rung.
2. Once you get to a rung, you can always climb to the next one up. (Notice that this assertion is an implication.)